

## Analog IC Design Design Challenge

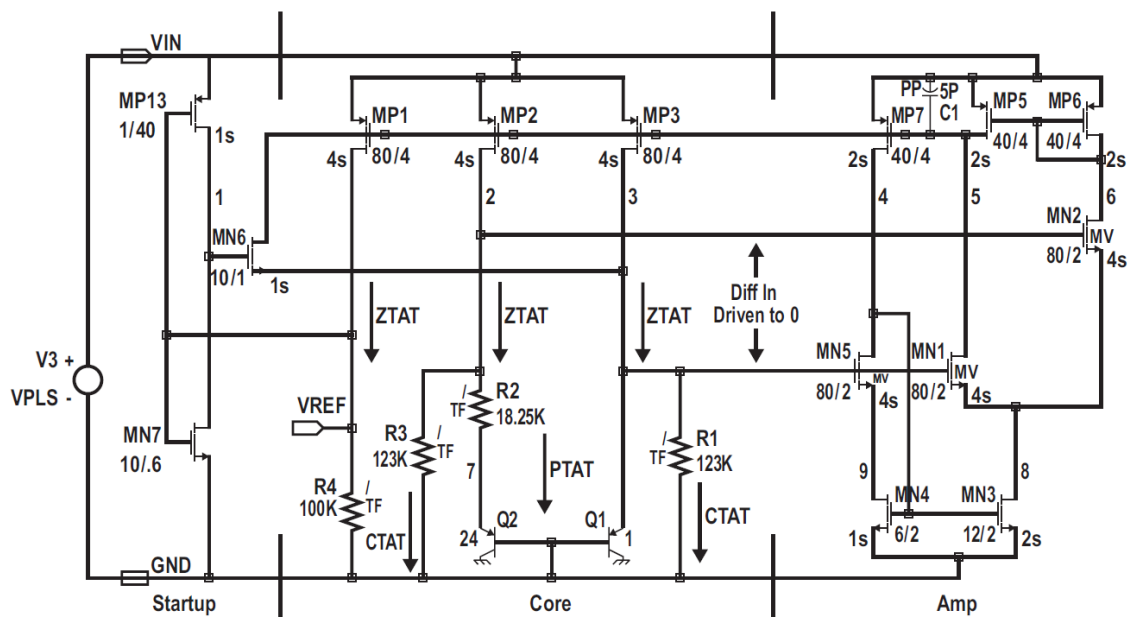
### Self-Biased Sub-1V Bandgap Reference Circuit

## Intended Learning Objectives

In this design challenge you will:

- Learn how to design a sub-1V self-biased BGR circuit.
- Learn how to analyze and redesign an existing circuit.
- Learn how to size an analog circuit on your own.
- Learn how variations affect the circuit.
- Learn how to design a startup circuit.

## Design Specs



It is required to redesign the BGR circuit shown above according to the following specifications.

|                    |                                             |
|--------------------|---------------------------------------------|
| Technology         | 65nm                                        |
| Supply voltage     | 1.2                                         |
| Corners            | Temp: -40 to 125<br>Process: SS, SF, FS, FF |
| Output voltage     | 0.8                                         |
| Total bias current | < 10uA                                      |
| Phase margin       | > 60°                                       |

# Deliverables

- Part 1 (6 pts): BGR core + startup circuit
  - Design the core BGR + startup circuit only
  - Replace the amplifier by a behavioral amplifier (VCVS)
  - Present the following:
    - Hand analysis
    - Schematics with device sizing.
    - Schematics with DC OP and node voltages annotated.
    - Vout vs temperature (-40 to 125) at TT, SS, FF, SF, FS
- Part 2 (4 pts): Replace the behavioral amplifier with the actual self-biased amplifier
  - Hint: You may use LVT transistors for the amplifier input pair
  - Present the following:
    - Hand analysis
    - Schematics with device sizing.
    - Schematics with DC OP and node voltages annotated.
    - Vout vs temperature (-40 to 125) at TT, SS, FF, SF, FS
    - STB analysis results
    - Transient simulation of Vout vs time
      - Apply VDD as a ramp from (0,0) to (1ms,VDD).
- Part 3 (2 pts – BONUS):
  - Replace the ideal passives (res/cap) with actual passives from the PDK. Tweak the circuit if needed.
  - Present the following:
    - The selected passives
    - Schematics with device sizing.
    - Schematics with DC OP and node voltages annotated.
    - Vout vs temperature (-40 to 125) at TT, SS, FF, SF, FS

# Acknowledgements

Thanks to all who contributed to these labs. If you find any errors or have suggestions concerning these labs, contact [Hesham.omran@eng.asu.edu.eg](mailto:Hesham.omran@eng.asu.edu.eg).